

# Planning Document: Total Salary Calculation by Role

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## Dataset and Task

The dataset consists of two JSON files: one representing employees and their roles (employees.json), and the other representing their salary payments for the months of January, February, and March 2024 (salary\_payments.json). The salary payments come from a live API as bulk JSON data. Figure 1 is a schema of the fields in the files and their relationship.

**Task:** Build a command-line application that calculates the total salary for each role in the company by month.

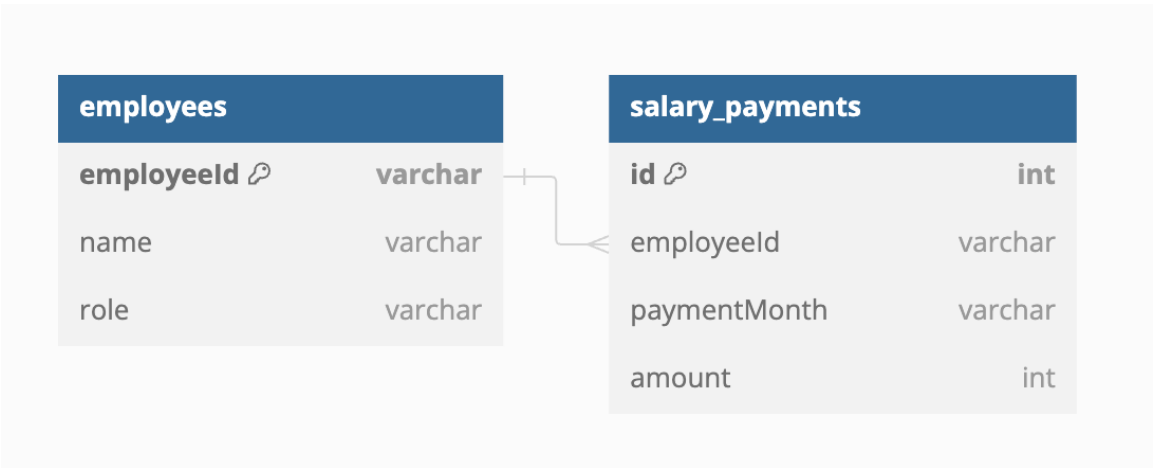


Figure 1: Employee Salaries Data Schema

## Important Fields and Joins

- employeeId: Key field for joining the Employees and Salary Payments datasets.
- role: Grouping field to calculate total salaries by role.
- paymentMonth: Field used to differentiate salary payments by month.
- amount: Field used to sum up the salaries.

## Steps to Perform the Task

To achieve the task, the following steps should be implemented:

1. Join the Employees and Salary Payments datasets on `employeeId`.
2. Group the data by `role` and `paymentMonth`.
3. Sum the `amount` for each group to calculate the total salary for each role by month.
4. Display the results in a tabular format with the columns Role and Total Salary, and rows representing each role and its corresponding total salary for each month.

## Reference

- ChatGPT 4o was used prompted to convert the two json files to the DBML format
- <https://dbdiagram.io/> was used to render the DBML format in a visual format (i.e. a schema diagram)